

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A.  $2.48 \times 10^{-8}$  cm
- B.  $5.92 \times 10^7$  cm/sec
- C.  $8.66 \times 10^{-6}$  cm
- D.  $3.14 \times 10^{-5}$  cm
- E.  $9.81 \times 10^{-4}$  cm
- F.  $5.27 \times 10^{-9}$  m
- G.  $1.23 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m
- I.  $6.625 \times 10^{-27}$  erg sec
- J.  $0.512 \times 10^6$  eV

Reasoning + Answer:

<think>  
 $KE = 1\text{ eV} = m_e v^2 / 2$   
 $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$   
 $m_e v^2 = 2 \times 1.6 \times 10^{-19}\text{ J}$   
 $v = \sqrt{2 \times 1.6 \times 10^{-19} / m_e}$   
 $h = 6.626 \times 10^{-34}\text{ Js}$ ,  $m_e = 9.11 \times 10^{-31}\text{ kg}$   
 $v = \sqrt{2 \times 1.6 \times 10^{-19} / 9.11 \times 10^{-31}}$   
 $v \approx 5.27 \times 10^{-9}\text{ m/s} \approx 5.27 \times 10^{-10}\text{ cm/s}$   
</think><answer>  
F.  $5.27 \times 10^{-9}\text{ m}$   
</answer>

